Global CO_2 Emissions, Economic Development, and Population Growth:

A Visual Analytics Approach

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Module: DSM050 Data Visualisation

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Declaration of use of generative Artificial Intelligence (AI) tools in the development of this assignment

During the preparation of this assignment, I used the following AI tools as described below. I have not copied and pasted material directly from any tool or source into my assignment. All writing has been reviewed, edited and finalised by me.

OpenAI (2026) ChatGPT (GPT-5.1): Used to improve structure, academic phrasing, proofreading, and clarity. All content was reviewed, edited and verified independently before submission.

Grammarly (2026). Grammarly: Used for proofreading, grammar correction, and sentence clarity.

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1. Introduction

Climate change is one of the most significant environmental challenges of the twenty-first century, with CO_2 emissions recognised as the primary driver of global warming. Continued industrialisation, economic growth and rising energy demand have substantially elevated atmospheric CO_2 concentrations, intensifying climate-related risks such as rising global temperatures, extreme weather events and sea-level rise (Friedlingstein et al., 2025). Understanding the factors that influence CO_2 emissions is increasingly important for informing evidence-based policy and long-term decarbonisation strategies.

Previous research identifies economic development, population growth and energy consumption as key drivers of CO_2 emissions, although the strength of these relationships varies across countries and stages of development. Early studies demonstrated that emissions generally increase alongside economic growth (Heil and Selden, 2001), while later research extended this by incorporating urbanisation, energy mix and demographic change (Gieraltowska et al., 2022). More recent literature has focused on the potential decoupling of economic growth from emissions, although evidence suggests that absolute decoupling remains limited (Haberl et al., 2020).

Despite these advances, global environmental datasets remain highly multidimensional, making it difficult to identify meaningful patterns through numerical summaries alone. Exploratory data visualisation complements statistical analysis by revealing temporal trends, cross-country differences and relationships between variables, enabling complex environmental data to be communicated more effectively (Kelleher and Wagener, 2011).

This study uses the *Our World in Data* CO_2 and Greenhouse Gas Emissions dataset to explore relationships between global CO_2 emissions, economic development and energy sources through exploratory data analysis and visualisation. The analysis aims to identify long-term emission trends, examine how economic development relates to emissions across countries, and investigate the contribution of different energy sources to national emission profiles. The findings are intended to support policymakers, environmental analysts and researchers by presenting complex environmental data in a clear and accessible manner.

2. Research Question and Literature Support

Understanding the drivers of CO_2 emissions is fundamental for developing effective climate mitigation strategies. While previous studies have established links between emissions, economic development and energy consumption, the complexity of global environmental datasets can make these relationships difficult to interpret (Heil and Selden, 2001). Exploratory data visualisation provides an effective approach for identifying trends, comparisons and interactions before more advanced statistical modelling is undertaken. Three research questions frame the analysis.

The first concerns how global CO_2 emissions have changed over time. Examining historical emission trends provides essential context for understanding the scale and progression of climate change. The Global Carbon Budget reports that global fossil CO_2 emissions have continued to increase despite international climate commitments, underscoring the ongoing challenge of reducing emissions (Friedlingstein et al., 2025). Earlier research similarly demonstrated that economic development has historically been accompanied by rising carbon output (Heil and Selden, 2001). Investigating long-term trends therefore helps identify periods of accelerated growth, temporary decline and overall shifts in global emission patterns.

The second question explores the relationship between economic development and CO_2 emissions. The Environmental Kuznets Curve suggests that emissions increase during early stages of development before declining as economies become wealthier and adopt cleaner technologies. However, empirical evidence remains mixed, with several studies arguing that economic growth alone does not guarantee lower emissions (Haberl et al., 2020). Visual analysis enables comparison of these relationships across countries, helping determine whether higher levels of economic development are consistently associated with greater emissions or whether alternative patterns emerge.

The third question examines how different energy sources contribute to emissions across countries. The composition of national energy systems plays a significant role in determining emission levels, and although renewable deployment has expanded globally, fossil fuels continue to dominate consumption (Halkos and Gkampoura, 2020). Moreover, countries differ considerably in their reliance on coal, oil, and gas, resulting in distinct emission profiles (Andrew, 2020). Comparing fuel-specific emissions across countries provides insight into how differences in national energy systems shape overall CO_2 emissions and highlights where future decarbonisation efforts may have the greatest potential impact.

3. Data Overview and Pre-Processing

3.1 Dataset

This study uses the Our World in Data (OWID) CO_2 and Greenhouse Gas Emissions dataset (Our World in Data, 2025), which combines information from the Global Carbon Project, the Energy Institute and several international statistical sources. The dataset contains annual observations for more than 200 countries and territories, covering the period from 1750 to 2024, and provides information on carbon dioxide emissions, as well as demographic, economic, and energy-related indicators (Friedlingstein et al., 2025).

Table 1: Overview of the dataset

Attribute	Description
Dataset	Our World in Data - CO_2 and Greenhouse Gas Emissions
Source	Our World in Data (compiled from the Global Carbon Project, Energy Institute, and other international sources)
Time Period	1750 - 2024
Coverage	215 countries and territories
Observations	23,408 (country-level, after cleaning)
Variables	17

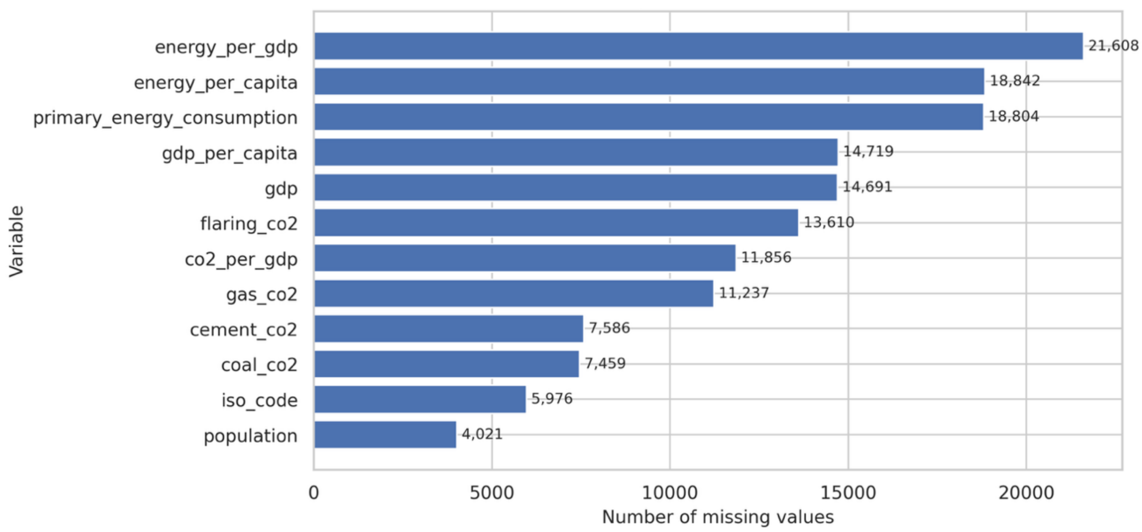
The dataset includes variables describing total CO_2 emissions, emissions per capita, gross domestic product (GDP), GDP per capita, population, and emissions from different energy sources, including coal, oil, gas, cement production, and gas flaring. These variables directly support the three research questions by enabling comparisons of historical emission trends, relationships between economic development and emissions, and the contribution of different energy sources. The dataset was selected because it offers consistent global coverage within a single harmonised file, avoiding the need to merge disparate sources (Andrew, 2020).

3.2 Data Cleaning

Prior to the analysis, the dataset was cleaned to improve consistency and ensure that the selected variables were suitable for visualisation. Only variables relevant to the research questions were retained, while unrelated greenhouse gas indicators and redundant measures were removed.

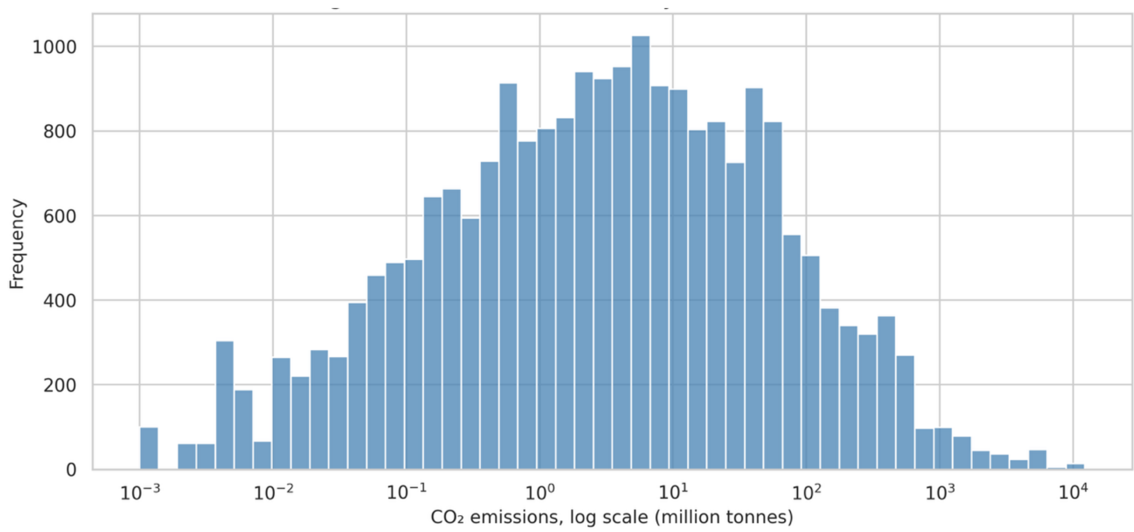
Missing values were assessed across the selected variables to determine their completeness before analysis. As illustrated in Figure 1, variables related to energy consumption and GDP contain the largest number of missing observations, particularly for earlier historical records where economic data are unavailable. Core variables such as total CO_2 emissions and population show substantially better coverage. Rather than imputing historical values, observations with missing values were excluded only where required for specific visualisations, preserving the integrity of the original dataset.

Figure 1. Missing values in selected variables.



The distribution of country-level CO_2 emissions was then examined. As shown in Figure 2, the data are highly positively skewed, with most countries producing relatively small amounts of CO_2 while a limited number of large economies account for extremely high emission levels. This wide range spans several orders of magnitude, making logarithmic scaling appropriate for subsequent visualisations involving emissions and economic indicators. Applying log transformations improves readability and allows countries with substantially different emission levels to be compared within the same figure.

Figure 2. Distribution of country-level CO_2 emissions (log scale).



Several variables were derived during pre-processing to support the analysis. GDP per capita was calculated by dividing total GDP by population, enabling fairer comparisons between countries of different sizes. A decade variable was also created to facilitate long-term trend analysis and reduce year-to-year fluctuations in the time-series visualisations.

Finally, relationships between the principal economic and environmental variables were explored using a Pearson correlation matrix. Figure 3 provides an overview of these associations and serves as an initial assessment of the dataset before detailed exploratory analysis. Strong positive correlations are observed between total CO_2 emissions and GDP, while GDP per capita and CO_2 emissions per capita also exhibit a moderate positive relationship. These results indicate that economic activity is closely associated with carbon emissions and provide a foundation for the more detailed analyses presented in Chapter 4.

4. Exploratory Data Analysis

Exploratory data analysis was conducted to investigate patterns, trends and relationships within the cleaned dataset through a series of visualisations. The analysis combines univariate and multivariate techniques to address the three research questions. Univariate visualisations were first used to examine the distributions of key variables, followed by comparative, time-series and relationship-based visualisations to identify trends and associations among CO_2 emissions, economic development and fossil fuel sources.

4.1 Global Emission Trends

Research Question 1: How have global CO_2 emissions changed over time, and which countries have contributed most to these changes?

Country coverage has increased considerably over time (Figure 4). Only a small number of countries contain records before the nineteenth century, whereas the dataset expands rapidly after the mid-twentieth century to include over 200 countries. This improved coverage should be considered when interpreting long-term global trends.

Despite the limited coverage in earlier years, Figure 5 demonstrates a clear long-term increase in global CO_2 emissions. Emissions remained relatively low until the Industrial Revolution, then accelerated throughout the twentieth century, particularly from the 1950s onwards. Although temporary declines are visible during major global events such as economic recessions and the COVID-19 pandemic, the overall upward trajectory remains consistent (Friedlingstein et al., 2025).

Figure 4. Country coverage over time.

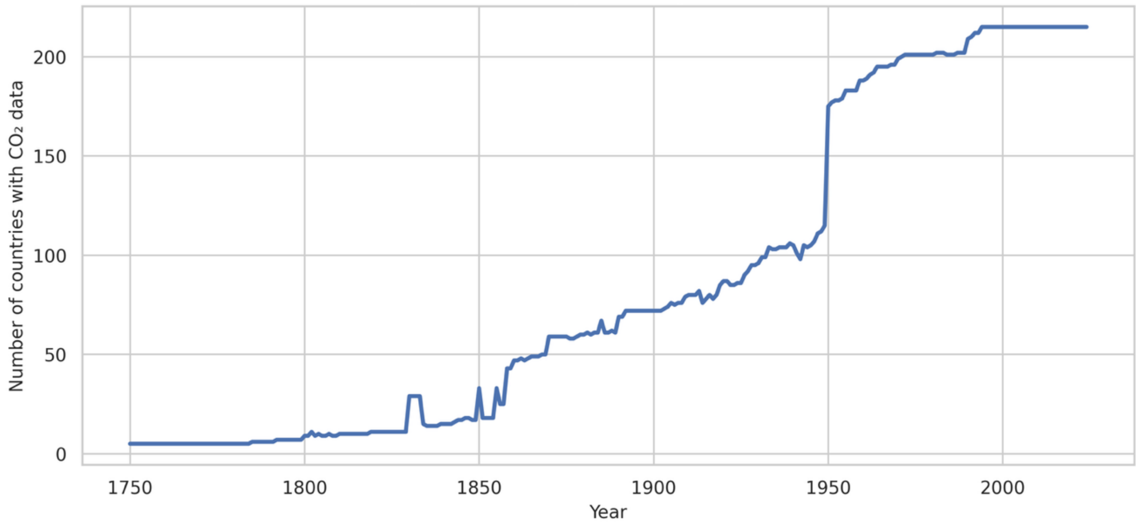


Figure 5. Global CO_2 emissions over time.

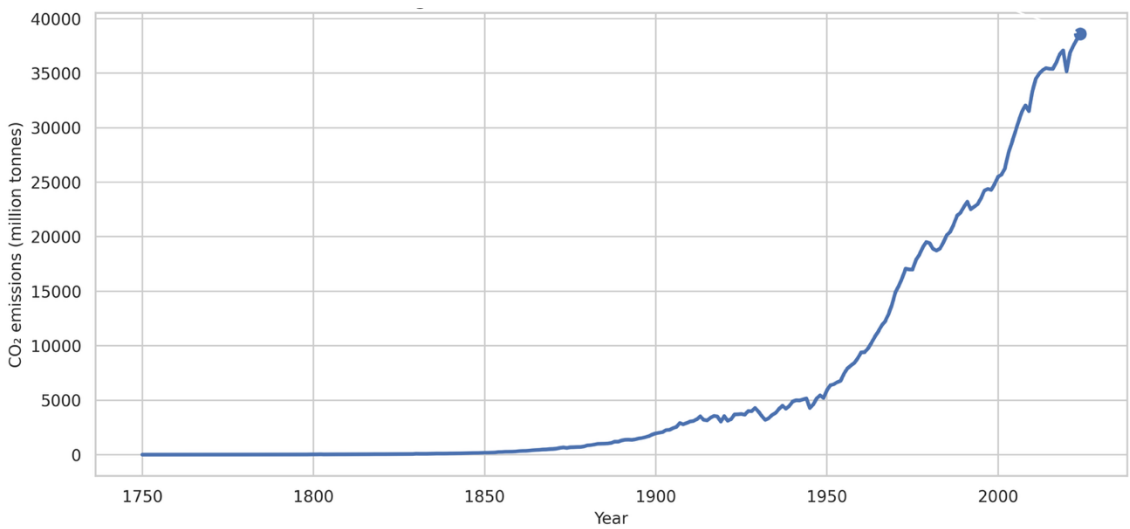


Figure 6 identifies the ten highest-emitting countries in the most recent year available, demonstrating that a relatively small number of countries contribute a disproportionately large share of global CO_2 emissions (Heil and Selden, 2001). Figure 7 extends this analysis by comparing historical emission trends among the five largest emitters, revealing differences in national trajectories and suggesting that economic growth, industrialisation and environmental policies influence long-term emission patterns. Overall, these findings answer RQ1 by demonstrating that global emissions have risen substantially over time and are increasingly concentrated among a small number of major economies.

Figure 6. Top ten CO_2-emitting countries (2024)

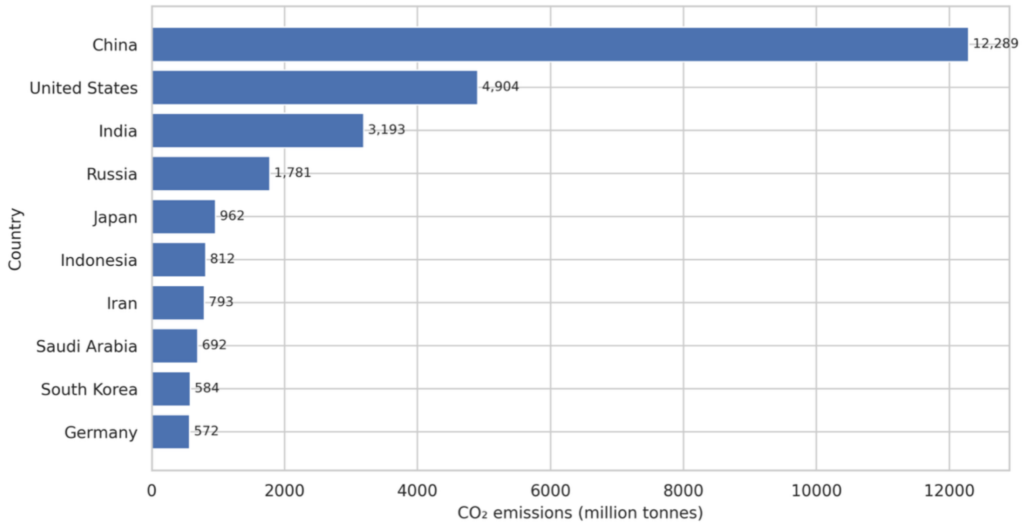
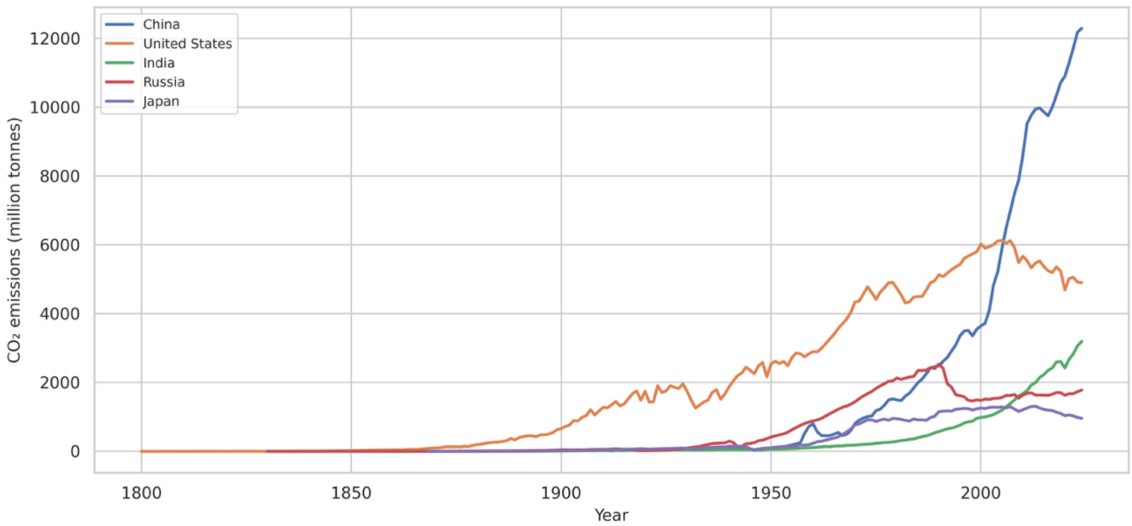


Figure 7. CO_2 emission trends for the five largest emitters.

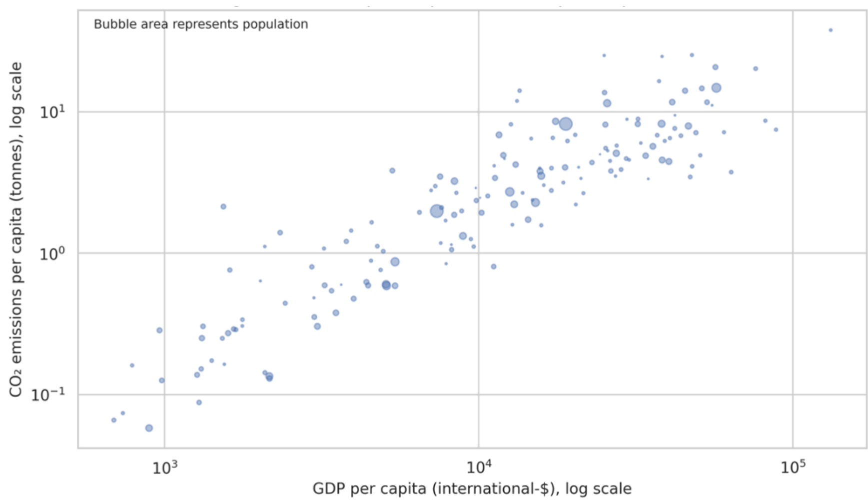


4.2 Economic Development and Emission

Research Question 2: What relationship exists between economic development and CO_2 emissions?

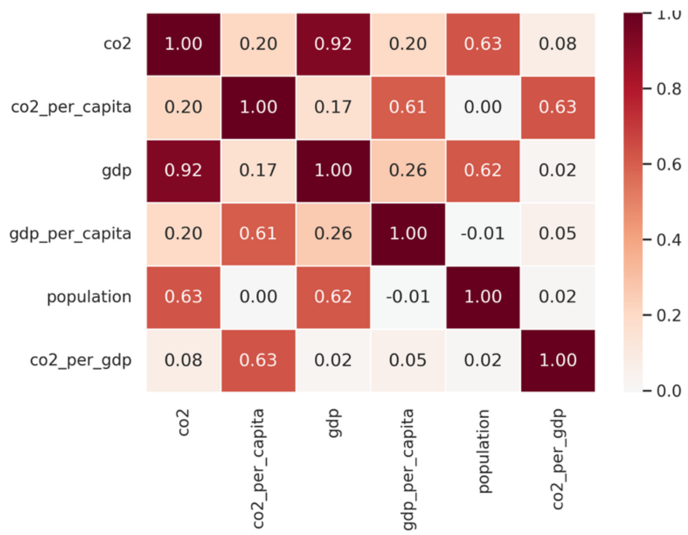
Figure 8 presents a bubble chart comparing GDP per capita with CO_2 emissions per capita, where bubble size represents national population. The visualisation reveals an overall positive relationship: countries with higher levels of economic development generally exhibit higher per-capita emissions. However, considerable variation exists across income levels, indicating that economic growth alone does not fully explain emission levels (Haberl et al., 2020). Several countries with similar GDP per capita display markedly different emission profiles, suggesting that energy mix, industrial structure and environmental policy also play important roles.

Figure 8. GDP per capita versus CO_2 emissions per capita (bubble chart).



The correlation matrix (Figure 9) quantitatively supports these findings. GDP and total CO_2 emissions show a strong positive association ($r = 0.92$), while the relationship between GDP per capita and CO_2 per capita is moderate ($r = 0.61$), and the correlation between population and total CO_2 is similarly moderate ($r = 0.63$). These results suggest that economic activity is closely associated with carbon emissions, although the moderate per-capita correlations indicate that economic development alone cannot fully account for national differences (Heil and Selden, 2001).

Figure 9. Correlation heatmap of economic and CO_2 indicators.



4.3 Energy Sources and Emissions

Research Question 3: How do different energy sources contribute to global CO_2 emissions?

Figure 10 presents a stacked area chart of global CO_2 emissions by energy source over time. Coal remains the dominant source across much of the historical period, while oil and gas have contributed increasingly since the mid-twentieth century. Cement and flaring account for smaller shares but remain relevant, as they show that emissions are linked not only to electricity and fuel use but also to industrial production (Halkos and Gkampoura, 2020).

Figure 10. Global CO_2 emissions by energy source.

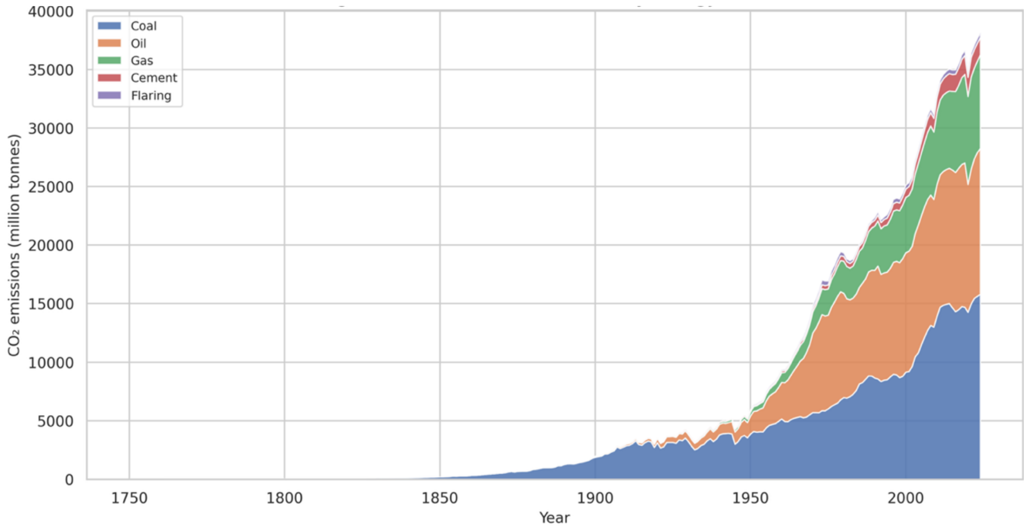


Figure 11 separates these sources into individual trend lines, making their growth patterns easier to compare. Coal, oil and gas all show sustained long-term growth, while cement and flaring remain comparatively smaller contributors. The relative contribution of natural gas has increased notably in recent decades, reflecting a shift in the global energy mix.

Figure 11. Trends in global CO_2 emissions by source.

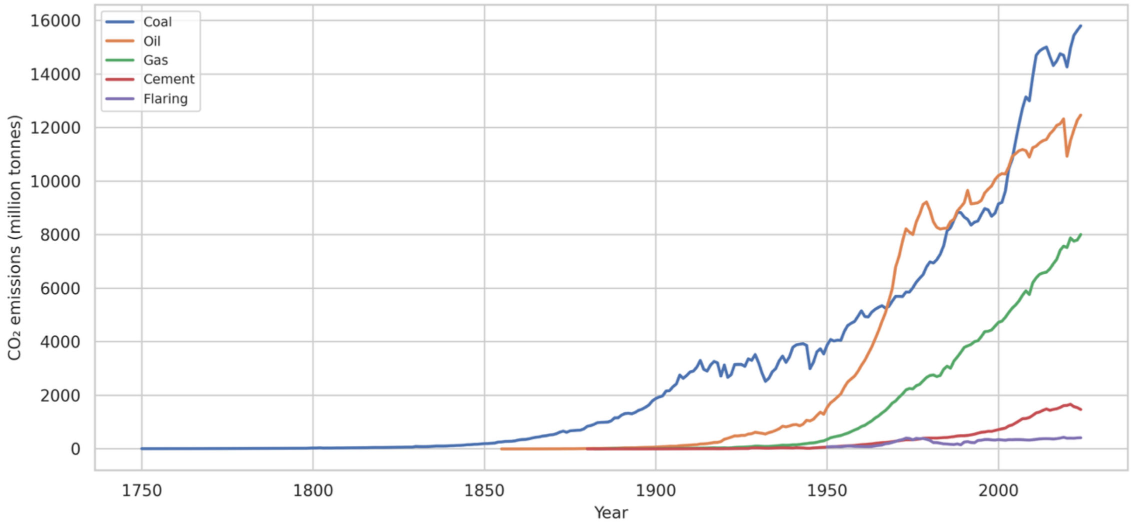


Figure 12 compares emissions from each source in the latest available year. Coal contributes the largest volume, followed by oil and natural gas. Figure 13 then shows how the composition of emissions has shifted over time using percentage shares by decade. Although fossil fuels continue to dominate, shifting proportions highlight the evolving roles of different energy sources in shaping global emission profiles (Andrew, 2020).

Figure 12. Global CO_2 emissions by source (2024).

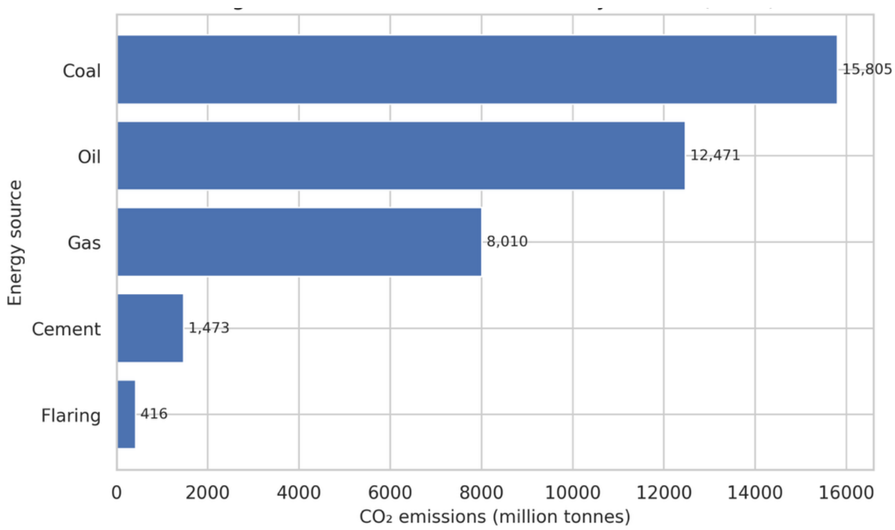
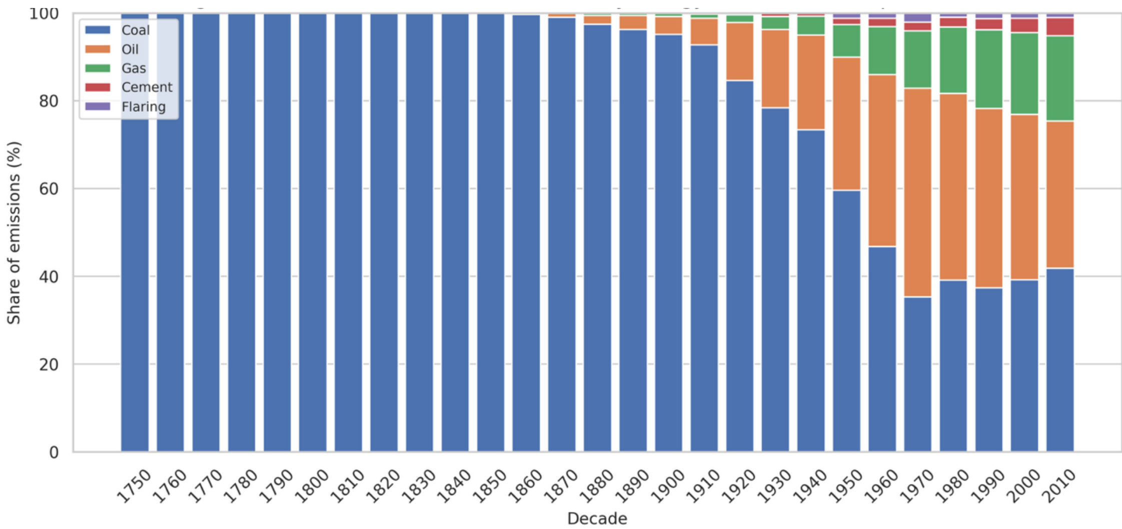


Figure 13. Share of global CO_2 emissions by energy source and decade.



4.4 Summary

Overall, the exploratory data analysis demonstrates clear differences in emission patterns across countries and confirms positive relationships between economic development, energy consumption and carbon emissions. Global CO_2 emissions have continued to increase, with a small number of countries accounting for a disproportionate share of total emissions. Economic development is associated with higher emissions, although the relationship varies substantially across countries, suggesting that energy mix, efficiency and policy also matter. Fossil fuels continue to dominate global emissions, underscoring the scale of the decarbonisation challenge. These findings provide the basis for the interpretation and discussion presented in the following section.

5. Interpretation, Implications and Limitations

5.1 Interpretation

Research Question 1: The analysis demonstrates that global CO_2 emissions have increased substantially over the past two and a half centuries, with a marked acceleration after the mid-twentieth century. Although temporary reductions are visible during periods of economic disruption, the overall trajectory remains firmly upward. The concentration of emissions among a relatively small number of countries indicates that global trends are driven primarily by large industrialised and rapidly developing economies rather than by uniform growth across all nations (Friedlingstein et al., 2025). These findings are consistent with previous research linking industrialisation and economic expansion to rising national emissions (Heil and Selden, 2001).

Research Question 2: The visual analysis indicates a generally positive relationship between economic development and CO_2 emissions, although considerable variation exists between countries with similar income levels. This suggests that economic growth alone does not determine emission levels. Instead, factors such as industrial structure, technological development and national energy policy appear to influence how efficiently economies generate wealth while limiting carbon emissions (Haberl et al., 2020). These observations broadly support previous research suggesting that economic growth and environmental sustainability are not inherently incompatible, but that successful decoupling requires deliberate policy intervention rather than occurring automatically.

Research Question 3: The analysis confirms that fossil fuels remain the dominant contributors to global CO_2 emissions, with coal, oil and natural gas accounting for the vast majority of total output throughout the study period (Halkos and Gkampoura, 2020). Regional differences in fuel composition demonstrate that countries face distinct decarbonisation challenges depending on their existing energy systems.

5.2 Implications

The findings carry several practical implications. Achieving meaningful global emissions reductions largely depends on accelerating decarbonisation in the major emitting economies, where international cooperation and coordinated climate policies are likely to deliver the greatest impact. Policymakers should invest in cleaner technologies and support low-carbon economic development rather than viewing growth and environmental protection as competing objectives. Effective climate strategies should also prioritise replacing existing fossil fuel infrastructure, particularly coal-fired power generation, while reflecting regional energy realities rather than adopting a universal approach.

5.3 Limitations

Several limitations should be acknowledged. First, historical observations contain missing values, particularly for GDP and energy-related variables, due to limited data availability in earlier years. Rather than estimating these values, the analysis relied only on available observations to preserve data accuracy. Second, this study is based entirely on observational data. While strong relationships between economic indicators and emissions are identified, these correlations should not be interpreted as evidence of causation. Many additional factors, including government policy, technological innovation and international trade, influence national emission levels but were not examined within this study. Finally, the analysis primarily uses descriptive visualisations to identify trends and patterns. Although these techniques effectively answer the research questions, they do not provide predictive capability. Future research could extend this work using statistical modelling or machine learning techniques to investigate the relative importance of different drivers of carbon emissions.

6. Evolution of Process and Personal Experience

At the start of this project, the Our World in Data CO_2 dataset felt overwhelming. With 79 variables and over 50,000 rows spanning nearly three centuries, the initial challenge was not writing code but deciding what to focus on. Selecting the 16 most relevant variables required repeatedly revisiting the research questions, and I learned that scoping is an analytical skill in itself rather than merely a preliminary step.

Early attempts at visualisation were often too busy to interpret. I initially tried to include every available variable in a single chart, which produced cluttered outputs that obscured rather than revealed patterns. Simplifying my approach, for example, using logarithmic scales for skewed emissions data and separating energy sources into individual line charts, taught me that effective visualisation is as much about what you remove as what you include. The correlation heatmap was a turning point: seeing the strong association between GDP and CO_2 emissions quantified numerically gave me confidence in the patterns I had observed visually and revealed that no single variable fully explains emission levels.

Working with missing data required me to think critically about the trade-offs between dataset size and data quality. Rather than imputing values, I chose complete-case analysis to prioritise reliability, a decision I had not considered in depth before. Overall, this project strengthened both my technical Python skills and my ability to reason analytically about data.

Word Count: 2468

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Appendix

Appendix A: Dataset Overview

Table 1: Overview of the dataset

Appendix B: Figures

Figure 1. Missing values in selected variables.
Horizontal bar chart showing the count of missing observations across each variable. Energy and GDP-related variables have the most gaps, particularly in early historical records, whereas core variables such as total CO_2 *emissions and population have substantially better coverage.*

Figure 2. Distribution of country-level CO_2 emissions (log scale).
Histogram of total CO_2 *emissions after log transformation. The original distribution is highly right-skewed; the log scale reveals the spread of emission levels across countries more clearly.*

Figure 3. Correlation matrix between economic and CO_2 indicators.
Heatmap displaying Pearson correlation coefficients between total CO_2 *emissions,* CO_2 *per capita, GDP, GDP per capita and population. Strong positive correlations are observed between total* CO_2 *and GDP (*$r = 0.92$*).*

Figure 4. Country coverage over time.
Line chart showing the number of countries with available CO_2 *records per year. Coverage is limited before the nineteenth century and expands rapidly after the mid-twentieth century to over 200 countries.*

Figure 5. Global CO_2 emissions over time.
Line chart of total global CO_2 *emissions from 1750 to the most recent year. Emissions remain low until the Industrial Revolution, accelerate from the 1950s onwards, and show temporary declines during major global events such as economic recessions and the COVID-19 pandemic.*

Figure 6. Top ten CO_2-emitting countries (2024)
Bar chart of the ten highest-emitting countries in the most recent year. A small number of large economies account for a disproportionately large share of global CO_2 *emissions.*

Figure 7. CO_2 emission trends for the five largest emitters.
Multi-line chart comparing historical CO_2 *emission trajectories of the five largest emitting countries. National trajectories differ, reflecting variations in industrialisation times, economic growth and environmental policy.*

Figure 8. GDP per capita versus CO_2 emissions per capita (bubble chart).
Scatter plot with bubble size representing national population. An overall positive relationship is evident, although considerable variation exists among countries with similar income levels, indicating that economic growth alone does not fully explain emissions.

Figure 9. Correlation heatmap of economic and CO_2 indicators.
Heatmap of Pearson correlation coefficients across all principal economic and environmental variables. GDP and total CO_2 *show a strong positive association (*$r = 0.92$*), while per-capita correlations are moderate (*$r = 0.61$*).*

Figure 10. Global CO_2 emissions by energy source.
Stacked area chart showing global CO_2 emissions decomposed by coal, oil, gas, cement and flaring over time. Coal remains the dominant source, with oil and gas contributing increasingly since the mid-twentieth century.

Figure 11. Trends in global CO_2 emissions by source.
Multi-line chart separating individual energy source trends. Coal, oil and gas all show sustained long-term growth, while cement and flaring remain smaller contributors. The share of natural gas has increased notably in recent decades.

Figure 12. Global CO_2 emissions by source (2024).
Horizontal bar chart comparing emissions from each energy source in the most recent year. Coal contributes the largest volume, followed by oil and natural gas, with cement and flaring considerably smaller.

Figure 13. Share of global CO_2 emissions by energy source and decade.
Percentage stacked bar chart showing how the composition of global emissions has shifted across decades. The changing proportions highlight the evolving role of different energy sources in the global emission profile.